

Group Project 2



■ 과 목 명 : 데이터베이스 전략 및 실습

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Database Design Report: ER Model Refinement and Relational Model Mapping

1. 개요

본 과제의 목적은 장비 운영 및 조달 관리 업무를 대상으로 ER 모델을 정교화하고, 이를 관계형 모델로 변환한 뒤 SQL DDL로 구현하는 것이다. 또한 생성된 테이블과 제약조건을 검토하여 데이터 무결성이 적절히 반영되었는지 확인하는 것을 목표로 한다. 본 설계는 business location, contract, machine, order, vendor, product, purchase request, purchase order, invoice 등의 핵심 엔터티를 중심으로 구성되었다.

2. ER Model Refinement

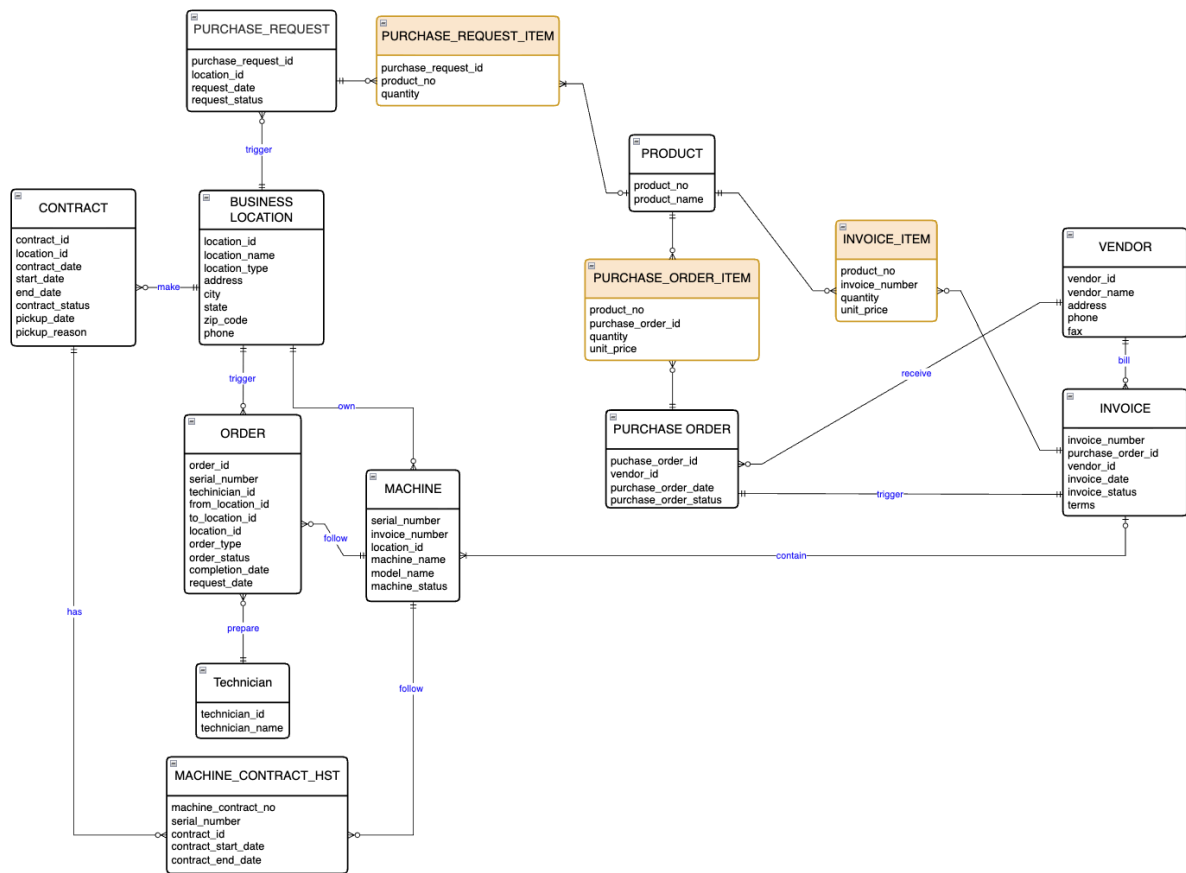
2.1 Main Entities

Entity	Description
business_location	창고, 매장, 현장 등 사업장 정보를 저장
contract	특정 사업장과의 계약 정보를 저장
machine	장비 기본 정보와 현재 배치 상태를 저장
machine_contract_hst	장비와 계약의 이력 관계를 저장
order	장비 이동/설치/수리 등 작업 오더를 저장
technician	작업 담당 기술자 정보를 저장
vendor	공급업체 정보를 저장
product	구매 대상 제품 정보를 저장
purchase_request	사업장의 구매 요청 정보를 저장
purchase_request_item	구매 요청 상세 품목을 저장
purchase_order	공급업체에 대한 구매 발주 정보를 저장
purchase_order_item	발주 상세 품목을 저장
invoice	공급업체 송장 정보를 저장
invoice_item	송장 상세 품목을 저장

2.2 Major Relationships

- 하나의 **business_location**은 여러 **contract**를 가질 수 있다.
- 하나의 **business_location**은 여러 **machine**을 보유할 수 있다.
- 하나의 **machine**은 여러 **order**에 참여할 수 있다.
- 하나의 **technician**은 여러 **order**를 처리할 수 있다.
- 하나의 **vendor**는 여러 **purchase_order**와 **invoice**를 가질 수 있다.
- 하나의 **purchase_order**는 여러 **purchase_order_item**을 가진다.
- 하나의 **invoice**는 여러 **invoice_item**을 가진다.
- **product**와 구매/송장 관련 테이블은 상세 테이블을 통해 연결된다.
- **machine_contract_hst**는 **machine**과 **contract** 간 이력성 관계를 표현한다.

3. 수정된 ER 모델



4. SQL DDL Script (Physical Model Implementation)

```

/* =====
DATABASE DDL SCRIPT
Equipment & Procurement Management System
MariaDB Version
===== */

/* =====
DROP TABLE SECTION
===== */

DROP TABLE IF EXISTS purchase_request_item;
DROP TABLE IF EXISTS purchase_order_item;
DROP TABLE IF EXISTS invoice_item;
DROP TABLE IF EXISTS machine_contract_hst;
DROP TABLE IF EXISTS `order`;
DROP TABLE IF EXISTS machine;
DROP TABLE IF EXISTS contract;
DROP TABLE IF EXISTS purchase_request;

```

```
DROP TABLE IF EXISTS invoice;
DROP TABLE IF EXISTS purchase_order;
DROP TABLE IF EXISTS technician;
DROP TABLE IF EXISTS product;
DROP TABLE IF EXISTS vendor;
DROP TABLE IF EXISTS business_location;
```

```
/* =====
MASTER TABLE SECTION
===== */
```

```
CREATE TABLE business_location (
location_id INT NOT NULL,
location_name VARCHAR(100) NOT NULL,
location_type VARCHAR(20) NOT NULL,
address VARCHAR(255),
city VARCHAR(255),
state VARCHAR(255),
zipcode VARCHAR(20),
phone VARCHAR(20),
PRIMARY KEY (location_id)
);
```

```
CREATE TABLE vendor (
vendor_id INT NOT NULL,
vendor_name VARCHAR(100) NOT NULL,
address VARCHAR(255) NOT NULL,
phone VARCHAR(20),
fax VARCHAR(20),
PRIMARY KEY (vendor_id)
);
```

```
CREATE TABLE product (
product_no INT NOT NULL,
product_name VARCHAR(100) NOT NULL,
PRIMARY KEY (product_no)
);
```

```
CREATE TABLE technician (
technician_id INT NOT NULL,
technician_name VARCHAR(100) NOT NULL,
PRIMARY KEY (technician_id)
);
```

```
/* =====  
TRANSACTION TABLE SECTION  
===== */
```

```
CREATE TABLE purchase_request (  
purchase_request_id INT NOT NULL,  
location_id INT NOT NULL,  
request_date DATETIME NOT NULL,  
request_status VARCHAR(20) NOT NULL,  
PRIMARY KEY (purchase_request_id),  
CONSTRAINT fk_business_location_to_purchase_request  
FOREIGN KEY (location_id)  
REFERENCES business_location(location_id)  
);
```

```
CREATE TABLE contract (  
contract_id INT NOT NULL,  
location_id INT NOT NULL,  
contract_date DATETIME NOT NULL,  
start_date DATETIME NOT NULL,  
end_date DATETIME NOT NULL,  
contract_status VARCHAR(20) NOT NULL,  
pickup_date DATETIME,  
pickup_reason VARCHAR(500),  
PRIMARY KEY (contract_id),  
CONSTRAINT fk_business_location_to_contract  
FOREIGN KEY (location_id)  
REFERENCES business_location(location_id)  
);
```

```
CREATE TABLE purchase_order (  
purchase_order_id INT NOT NULL,  
vendor_id INT NOT NULL,  
purchase_order_date DATETIME NOT NULL,  
purchase_order_status VARCHAR(20) NOT NULL,  
PRIMARY KEY (purchase_order_id),  
CONSTRAINT fk_purchase_order  
FOREIGN KEY (vendor_id)  
REFERENCES vendor(vendor_id)  
);
```

```
CREATE TABLE invoice (  
invoice_number INT NOT NULL,  
purchase_order_id INT NOT NULL,  
vendor_id INT NOT NULL,
```

```
invoice_date DATETIME NOT NULL,  
invoice_status VARCHAR(20) NOT NULL,  
terms VARCHAR(500) NOT NULL,  
PRIMARY KEY (invoice_number),  
CONSTRAINT fk_purchase_order_to_invoice  
FOREIGN KEY (purchase_order_id)  
REFERENCES purchase_order(purchase_order_id),  
CONSTRAINT fk_vendor_to_invoice  
FOREIGN KEY (vendor_id)  
REFERENCES vendor(vendor_id)  
);
```

```
CREATE TABLE machine (  
serial_number INT NOT NULL,  
invoice_number INT NOT NULL,  
location_id INT,  
machine_name VARCHAR(100) NOT NULL,  
model_name VARCHAR(100) NOT NULL,  
machine_status VARCHAR(20) NOT NULL,  
PRIMARY KEY (serial_number),  
CONSTRAINT fk_business_location_to_machine  
FOREIGN KEY (location_id)  
REFERENCES business_location(location_id),  
CONSTRAINT fk_invoice_to_machine  
FOREIGN KEY (invoice_number)  
REFERENCES invoice(invoice_number)  
);
```

```
CREATE TABLE machine_contract_hst (  
machine_contract_no INT NOT NULL,  
serial_number INT NOT NULL,  
contract_id INT NOT NULL,  
contract_start_date DATETIME NOT NULL,  
contract_end_date DATETIME NOT NULL,  
PRIMARY KEY (machine_contract_no),  
CONSTRAINT fk_machine_to_machine_contract_hst  
FOREIGN KEY (serial_number)  
REFERENCES machine(serial_number),  
CONSTRAINT fk_contract_to_machine_contract_hst  
FOREIGN KEY (contract_id)  
REFERENCES contract(contract_id)  
);
```

```
CREATE TABLE `order` (  
order_id INT NOT NULL,
```

```

serial_number INT NOT NULL,
technician_id INT,
from_location_id INT,
to_location_id INT NOT NULL,
location_id INT,
order_type VARCHAR(20) NOT NULL,
order_status VARCHAR(20) NOT NULL,
completion_date DATETIME,
request_date DATETIME NOT NULL,
PRIMARY KEY (order_id),
CONSTRAINT fk_machine_to_order
FOREIGN KEY (serial_number)
REFERENCES machine(serial_number),
CONSTRAINT fk_Techinician_to_order
FOREIGN KEY (technician_id)
REFERENCES technician(technician_id),
CONSTRAINT fk_business_location_to_order
FOREIGN KEY (location_id)
REFERENCES business_location(location_id),
CONSTRAINT fk_business_location_to_order2
FOREIGN KEY (from_location_id)
REFERENCES business_location(location_id),
CONSTRAINT fk_business_location_to_order3
FOREIGN KEY (to_location_id)
REFERENCES business_location(location_id)
);

```

```

/* =====
DETAIL TABLE SECTION (RESOLVE M:N RELATIONSHIPS)
===== */

```

```

CREATE TABLE purchase_request_item (
purchase_request_id INT NOT NULL,
product_no INT NOT NULL,
quantity INT NOT NULL,
PRIMARY KEY (purchase_request_id, product_no),
CONSTRAINT fk_product_to_purchase_request_item
FOREIGN KEY (product_no)
REFERENCES product(product_no),
CONSTRAINT fk_purchase_request_to_purchase_request_item
FOREIGN KEY (purchase_request_id)
REFERENCES purchase_request(purchase_request_id)
);

```

```

CREATE TABLE purchase_order_item (
product_no INT NOT NULL,

```



```
purchase_order_id INT NOT NULL,  
quantity INT NOT NULL,  
unit_price DECIMAL(18,2) NOT NULL,  
PRIMARY KEY (product_no, purchase_order_id),  
CONSTRAINT fk_product_to_purchase_order_item  
FOREIGN KEY (product_no)  
REFERENCES product(product_no),  
CONSTRAINT fk_purchase_order_to_purchase_order_item  
FOREIGN KEY (purchase_order_id)  
REFERENCES purchase_order(purchase_order_id)  
);
```

```
CREATE TABLE invoice_item (  
product_no INT NOT NULL,  
invoice_number INT NOT NULL,  
quantity INT NOT NULL,  
unit_price DECIMAL(18,2) NOT NULL,  
PRIMARY KEY (product_no, invoice_number),  
CONSTRAINT fk_product_to_invoice_item  
FOREIGN KEY (product_no)  
REFERENCES product(product_no),  
CONSTRAINT fk_invoice_to_invoice_item  
FOREIGN KEY (invoice_number)  
REFERENCES invoice(invoice_number)  
);
```

5. 생성된 테이블 검증 (Table Verification)

```
/* =====  
전체 테이블 목록 조회  
===== */
```

```
SELECT table_name  
  
FROM information_schema.tables  
  
WHERE table_schema = 'mng_db';
```

```
table_name      |  
-----+  
invoice         |  
purchase_request|  
vendor          |  
invoice_item    |  
machine         |  
purchase_order_item|  
business_location|  
product        |  
contract       |  
machine_contract_hst|  
purchase_order  |  
purchase_request_item|  
technician     |  
order          |
```

```
/* =====  
각 테이블 컬럼(속성) 확인  
===== */
```

desc business_location;

Field	Type	Null	Key	Default	Extra
location_id	int(11)	NO	PRI		
location_name	varchar(100)	NO			
location_type	varchar(20)	NO			
address	varchar(255)	YES			
city	varchar(255)	YES			
state	varchar(255)	YES			
zipcode	varchar(20)	YES			
phone	varchar(20)	YES			

desc contract;

Field	Type	Null	Key	Default	Extra
contract_id	int(11)	NO	PRI		
location_id	int(11)	NO	MUL		
contract_date	datetime	NO			
start_date	datetime	NO			
end_date	datetime	NO			
contract_status	varchar(20)	NO			
pickup_date	datetime	YES			
pickup_reason	varchar(500)	YES			

desc invoice;

Field	Type	Null	Key	Default	Extra
-----+-----+---+---+-----+-----+					
invoice_number	int(11)	NO	PRI		
purchase_order_id	int(11)	NO	MUL		
vendor_id	int(11)	NO	MUL		
invoice_date	datetime	NO			
invoice_status	varchar(20)	NO			
terms	varchar(500)	NO			

desc invoice_item;

Field	Type	Null	Key	Default	Extra
-----+-----+---+---+-----+-----+					
product_no	int(11)	NO	PRI		
invoice_number	int(11)	NO	PRI		
quantity	int(11)	NO			
unit_price	decimal(18,2)	NO			

desc machine;

Field	Type	Null	Key	Default	Extra
-----+-----+---+---+-----+-----+					
serial_number	int(11)	NO	PRI		
invoice_number	int(11)	NO	MUL		
location_id	int(11)	YES	MUL		
machine_name	varchar(100)	NO			

```

model_name      |varchar(100)|NO  |  |      |      |
machine_status |varchar(20) |NO  |  |      |      |

```

desc machine_contract_hst;

```

Field            |Type      |Null|Key|Default|Extra|
-----+-----+----+---+-----+-----+
machine_contract_no|int(11)   |NO  |PRI|        |      |
serial_number     |int(11)   |NO  |MUL|        |      |
contract_id       |int(11)   |NO  |MUL|        |      |
contract_start_date|datetime  |NO  |   |        |      |
contract_end_date  |datetime  |NO  |   |        |      |

```

desc `order`;

```

Field            |Type      |Null|Key|Default|Extra|
-----+-----+----+---+-----+-----+
order_id         |int(11)   |NO  |PRI|        |      |
serial_number     |int(11)   |NO  |MUL|        |      |
technician_id    |int(11)   |YES  |MUL|        |      |
from_location_id |int(11)   |YES  |MUL|        |      |
to_location_id   |int(11)   |NO  |MUL|        |      |
location_id      |int(11)   |YES  |MUL|        |      |
order_type       |varchar(20)|NO  |   |        |      |
order_status     |varchar(20)|NO  |   |        |      |
completion_date  |datetime  |YES  |   |        |      |
request_date     |datetime  |NO  |   |        |      |

```

desc product;

Field	Type	Null	Key	Default	Extra
product_no	int(11)	NO	PRI		
product_name	varchar(100)	NO			

desc purchase_order;

Field	Type	Null	Key	Default	Extra
purchase_order_id	int(11)	NO	PRI		
vendor_id	int(11)	NO	MUL		
purchase_order_date	datetime	NO			
purchase_order_status	varchar(20)	NO			

desc purchase_order_item;

Field	Type	Null	Key	Default	Extra
product_no	int(11)	NO	PRI		
purchase_order_id	int(11)	NO	PRI		
quantity	int(11)	NO			
unit_price	decimal(18,2)	NO			

desc purchase_request;

Field	Type	Null	Key	Default	Extra
purchase_request_id	int(11)	NO	PRI		
location_id	int(11)	NO	MUL		
request_date	datetime	NO			
request_status	varchar(20)	NO			

desc purchase_request_item;

Field	Type	Null	Key	Default	Extra
purchase_request_id	int(11)	NO	PRI		
product_no	int(11)	NO	PRI		
quantity	int(11)	NO			

desc technician;

Field	Type	Null	Key	Default	Extra
technician id	int(11)	NO	PRI		
technician_name	varchar(100)	NO			

desc vendor;

Field	Type	Null	Key	Default	Extra
vendor_id	int(11)	NO	PRI		

vendor_name	varchar(100) NO				
address	varchar(255) NO				
phone	varchar(20) YES				
fax	varchar(20) YES				

```
/* =====
```

테이블 구조 확인

```
===== */
```

SELECT

```

    table_name,
    column_name,
    column_type,
    is_nullable,
    column_key,
    column_default,
    extra

```

FROM information_schema.columns

WHERE table_schema = 'mng_db'

ORDER BY table_name, ordinal_position;

table_name	column_name	column_type	is_nullable	column_key	column
business_location	location_id	int(11)	NO	PRI	
business_location	location_name	varchar(100)	NO		
business_location	location_type	varchar(20)	NO		
business_location	address	varchar(255)	YES		NULL
business_location	city	varchar(255)	YES		NULL
business_location	state	varchar(255)	YES		NULL
business_location	zipcode	varchar(20)	YES		NULL
business_location	phone	varchar(20)	YES		NULL
contract	contract_id	int(11)	NO	PRI	
contract	location_id	int(11)	NO	MUL	
contract	contract_date	datetime	NO		
contract	start_date	datetime	NO		
contract	end_date	datetime	NO		
contract	contract_status	varchar(20)	NO		
contract	pickup_date	datetime	YES		NULL
contract	pickup_reason	varchar(500)	YES		NULL
invoice	invoice_number	int(11)	NO	PRI	
invoice	purchase_order_id	int(11)	NO	MUL	
invoice	vendor_id	int(11)	NO	MUL	
invoice	invoice_date	datetime	NO		
invoice	invoice_status	varchar(20)	NO		
invoice	terms	varchar(500)	NO		
invoice_item	product_no	int(11)	NO	PRI	
invoice_item	invoice_number	int(11)	NO	PRI	
invoice_item	quantity	int(11)	NO		
invoice_item	unit_price	decimal(18,2)	NO		
machine	serial_number	int(11)	NO	PRI	
machine	invoice_number	int(11)	NO	MUL	
machine	location_id	int(11)	YES	MUL	NULL
machine	machine_name	varchar(100)	NO		
machine	model_name	varchar(100)	NO		
machine	machine_status	varchar(20)	NO		

machine_contract_hst	machine_contract_no	int(11)	NO	PRI	
machine_contract_hst	serial_number	int(11)	NO	MUL	
machine_contract_hst	contract_id	int(11)	NO	MUL	
machine_contract_hst	contract_start_date	datetime	NO		
machine_contract_hst	contract_end_date	datetime	NO		
order	order_id	int(11)	NO	PRI	
order	serial_number	int(11)	NO	MUL	
order	technician_id	int(11)	YES	MUL	NULL
order	from_location_id	int(11)	YES	MUL	NULL
order	to_location_id	int(11)	NO	MUL	
order	location_id	int(11)	YES	MUL	NULL
order	order_type	varchar(20)	NO		
order	order_status	varchar(20)	NO		
order	completion_date	datetime	YES		NULL
order	request_date	datetime	NO		
product	product_no	int(11)	NO	PRI	
product	product_name	varchar(100)	NO		
purchase_order	purchase_order_id	int(11)	NO	PRI	
purchase_order	vendor_id	int(11)	NO	MUL	
purchase_order	purchase_order_date	datetime	NO		
purchase_order	purchase_order_status	varchar(20)	NO		
purchase_order_item	product_no	int(11)	NO	PRI	
purchase_order_item	purchase_order_id	int(11)	NO	PRI	
purchase_order_item	quantity	int(11)	NO		
purchase_order_item	unit_price	decimal(18,2)	NO		
purchase_request	purchase_request_id	int(11)	NO	PRI	
purchase_request	location_id	int(11)	NO	MUL	
purchase_request	request_date	datetime	NO		
purchase_request	request_status	varchar(20)	NO		
purchase_request_item	purchase_request_id	int(11)	NO	PRI	
purchase_request_item	product_no	int(11)	NO	PRI	
purchase_request_item	quantity	int(11)	NO		
technician	technician_id	int(11)	NO	PRI	
technician	technician_name	varchar(100)	NO		
vendor	vendor_id	int(11)	NO	PRI	
vendor	vendor_name	varchar(100)	NO		
vendor	address	varchar(255)	NO		
vendor	phone	varchar(20)	YES		NULL
vendor	fax	varchar(20)	YES		NULL

```
/* =====
```

제약조건 확인

```
===== */
```

SELECT

tc.constraint_name,

tc.constraint_type,

tc.table_name

FROM information_schema.table_constraints tc

WHERE tc.table_schema = 'mng_db';

constraint_name	constraint_type	table_name
-----+-----+-----		
PRIMARY	PRIMARY KEY	invoice
fk_purchase_order_to_invoice	FOREIGN KEY	invoice
fk_vendor_to_invoice	FOREIGN KEY	invoice
PRIMARY	PRIMARY KEY	purchase_request
fk_business_location_to_purchase_request	FOREIGN KEY	purchase_request
PRIMARY	PRIMARY KEY	vendor
PRIMARY	PRIMARY KEY	invoice_item
fk_invoice_to_Invoice_item	FOREIGN KEY	invoice_item
fk_product_to_invoice_item	FOREIGN KEY	invoice_item
PRIMARY	PRIMARY KEY	machine
fk_business_location_to_machine	FOREIGN KEY	machine
fk_invoice_to_machine	FOREIGN KEY	machine
PRIMARY	PRIMARY KEY	purchase_order_item
fk_product_to_purchase_order_item	FOREIGN KEY	purchase_order_item
fk_purchase_order_to_purchase_order_item	FOREIGN KEY	purchase_order_item
PRIMARY	PRIMARY KEY	business_location
PRIMARY	PRIMARY KEY	product
PRIMARY	PRIMARY KEY	contract
fk_business_location_to_contract	FOREIGN KEY	contract
PRIMARY	PRIMARY KEY	machine_contract_hst
fk_contract_to_machine_contract_hst	FOREIGN KEY	machine_contract_hst
fk_machine_to_machine_contract_hst	FOREIGN KEY	machine_contract_hst
PRIMARY	PRIMARY KEY	purchase_order

fk_purchase_order	FOREIGN KEY	purchase_order	
PRIMARY	PRIMARY KEY	purchase_request_item	
fk_product_to_purchase_request_item	FOREIGN KEY	purchase_request_item	
fk_purchase_request_to_purchase_request_item	FOREIGN KEY	purchase_request_item	
PRIMARY	PRIMARY KEY	technician	
PRIMARY	PRIMARY KEY	order	
fk_Technician_to_order	FOREIGN KEY	order	
fk_business_location_to_order	FOREIGN KEY	order	
fk_business_location_to_order2	FOREIGN KEY	order	
fk_business_location_to_order3	FOREIGN KEY	order	
fk_machine_to_order	FOREIGN KEY	order	